

Citation Faithfulness in Open-Weight Retrieval-Augmented Generation

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ABSTRACT

blue Citation correctness does not establish that a retrieval-augmented generation system used the cited document. We evaluate this distinction in Llama-3.1-8B-Instruct, Qwen2.5-7B-Instruct, and Gemma-3-12B-it using 1,444 Natural Questions queries with KILT Wikipedia retrieval, 2,000 ConflictBank examples per model, a Llama-only mechanistic set of 31 PopQA capital-city pairs, and hidden-state probes. When the answer phrase was inserted into forged documents, random documents were rarely cited, whereas relevant but originally uncited documents reached 9.7% for Qwen and cited-other documents reached 21.6% for Qwen, 17.8% for Gemma, and 9.1% for Llama. Under a string-matching classifier that often marked closed-book answers as ambiguous, supplied conflict evidence increased context matches. In the observed test samples, probes achieved AUROC values of 0.943, 0.969, and 0.831 for Llama, Qwen, and Gemma, respectively; the small positive counts make these estimates especially uncertain. The mechanistic localization found no positive components at the prespecified threshold; scaling selected components corrected 0/7 missed-citation and 0/58 spurious-citation cases. The results show that behavioral post-rationalization is measurable, context-sensitive, and partly decodable from internal states, while the tested causal intervention was insufficient for generation-level control.

1. Introduction

Retrieval-augmented generation (RAG) combines language models with retrieved documents Lewis, Perez, Piktus, Karpukhin, Goyal, Kuttler, Lewis, Yih, Rocktaschel, Riedel and Kiela (2020); Guu, Lee, Tung, Pasupat and Chang (2020); Izacard and Grave (2021). It is useful because models cannot store every relevant fact in their weights, their knowledge can become outdated, and they can produce plausible but unsupported answers Mallen, Asai, Zhong, Das, Khashabi and Hajishirzi (2023). Retrieval supplies external documents that may be more current, specific, and easier to verify than stored knowledge. In practical RAG systems, citations connect an answer to its claimed evidence, allowing readers to inspect the source of a claim; they are therefore often treated as signs that the cited material supported the answer, particularly in information-seeking settings where users rely on citations to assess trustworthiness Liu, Zhang and Liang (2023).

A cited answer can look transparent while hiding several separate steps: the retriever selects documents, the generator writes an answer using both those documents and stored knowledge, and the system adds a citation marker pointing to one of the retrieved sources. Performing the final step well does not show that the earlier steps worked as intended. A system may retrieve a useful passage, ignore it, answer from memory, and then attach a citation that happens to match the answer, leaving readers with an output that appears grounded but does not explain how the answer was produced Wallat, Heuss, de Rijke and Anand (2025). Citations are therefore more than formatting; they serve as practical signals of provenance, auditability, and reliance on external evidence.

Producing citations is not the same as proving that they reflect what the model used. Attributed-generation systems have made progress in writing answers with citations and checking whether cited passages support individual claims Gao, Yen, Yu and Chen (2023); Liu et al. (2023); however, these checks mainly measure citation correctness or completeness, asking whether a passage supports a claim and whether important claims have support, rather than whether the model used that passage instead of relying on stored knowledge or a shallow document-answer match. A citation can therefore be correct in content while still providing a misleading account of the model's evidence use.

We define *citation correctness* as whether a cited passage supports the claim made in the generated answer, whereas *citation faithfulness* asks whether that passage corresponds to evidence that actually influenced the model during generation. Correctness is a post-generation textual relation; faithfulness concerns evidence use and is therefore a causal question that requires testing how generation changes when the evidence changes. No single test answers this perfectly:

ORCID(s):

behavioral interventions assess sensitivity to forged support, activation patching tests specific internal mechanisms, and probes reveal whether a risk signal is present in hidden states. Because these methods answer different questions, we treat them as complementary.

Wallat et al. formalize this distinction for RAG attributions and introduce post-rationalization interventions Wallat et al. (2025), in which an answer-bearing string is inserted into a document that was not the original factual source. When a model repeats the target statement and cites that document, the citation is difficult to interpret as true provenance. This approach connects with work that studies attribution from inside the model through context-sensitive token analysis and saliency-based document attribution Qi, Sarti, Fernández and Bisazza (2024), motivating evaluations that separate textual support from actual reliance on retrieved context.

Open-weight instruction-tuned models are useful for this study because they support controlled generation experiments across model families while providing access to hidden states and intermediate activations. We can therefore compare how models behave under document interventions, what information their representations contain, and whether selected internal components can change citation behavior Leon (2025). Access to activations does not, however, make faithfulness directly visible: a good probe demonstrates predictability, a successful patch demonstrates a causal effect under one intervention, and neither alone proves that a citation fully explains the model’s reasoning.

The Research Objectives section formalizes these questions: behavioral sensitivity to forged evidence, context sensitivity under knowledge conflict, and the predictability and causal control of citation behavior.

This paper studies citation faithfulness in three open-weight instruction-tuned RAG models: Llama-3.1-8B-Instruct, Qwen2.5-7B-Instruct, and Gemma-3-12B-it. We make four contributions. First, we measure post-rationalization on Natural Questions with KILT Wikipedia retrieval under random, relevant-uncited, and cited-other document interventions. Second, we use ConflictBank to study what models do when supplied context conflicts with stored knowledge. Third, we test a focused causal hypothesis about Llama-3.1-8B-Instruct with activation patching on PopQA capital-city examples. Fourth, we test whether post-rationalization labels can be decoded from hidden states, and whether these probes beat lexical, logit-margin, and embedding-similarity baselines. The theoretical contribution is a framework that distinguishes citation correctness, behavioral evidence reliance, representation-level predictability, and causal control, thereby clarifying how this study differs from work that evaluates only whether a cited passage supports a claim. The practical contribution is an evaluation and monitoring strategy that combines intervention-based faithfulness tests with calibrated hidden-state screening, helping RAG developers identify citations that are supportable but may not reflect the evidence used during generation. Together, these experiments separate behavior, causal mechanisms, and prediction without treating any one of them as the same thing as citation correctness.

The rest of the paper is organized as follows. Section 3 reviews work on RAG, attributed generation, knowledge conflicts, mechanistic interpretability, and probing. Section 4 describes the datasets and model checkpoints, while Section 5 describes the research architecture and experimental methods. Section 6 presents the behavioral, conflict, mechanistic, and probe results. Section 7 discusses what the results mean for citation faithfulness. Sections 8 and 9 describe limitations and conclusions.

2. Research Objectives

The objective of this study is to determine whether citations in open-weight RAG systems reflect the evidence used during generation, rather than merely evidence that supports the final answer. We address this objective through three research questions:

1. **RQ1:** How often do models recover an answer-bearing phrase from a forged document and cite that document, and how does the rate vary with document relevance and citation context?
2. **RQ2:** When supplied evidence conflicts with stored knowledge, how strongly do the models follow the supplied context?
3. **RQ3:** Can post-rationalization be predicted from hidden states, and can activation interventions change citation behavior during full generation?

These questions separate three claims that are often conflated: behavioral sensitivity to an intervention, predictive information in model representations, and causal control of generation. The study therefore treats citation correctness as a necessary but insufficient property of citation faithfulness.

3. Related Work

Early RAG systems joined language models with external retrieval for knowledge-intensive tasks. REALM adds retrieval during language-model pre-training. RAG generates while marginalizing over retrieved passages. Fusion-in-Decoder lets the decoder combine evidence from several passages Guu et al. (2020); Lewis et al. (2020); Izacard and Grave (2021). These systems showed that retrieval can improve factual and open-domain question answering. However, the usual metrics mainly measure answer quality. Natural Questions provides real user questions with annotated answers, and KILT gives a shared Wikipedia source with provenance annotations Kwiatkowski, Palomaki, Redfield, Collins, Parikh, Alberti, Epstein, Polosukhin, Devlin, Lee, Toutanova, Jones, Kelcey, Chang, Dai, Uszkoreit, Le and Petrov (2019); Petroni, Piktus, Fan, Lewis, Yazdani, De Cao, Thorne, Jernite, Karpukhin, Maillard, Plachouras, Rocktaschel and Riedel (2021). These datasets make retrieval and source tracking easier to evaluate. Still, a relevant passage and a correct answer do not prove that the generator relied on that passage for a specific claim.

Attributed generation makes sources visible by asking systems to attach references to generated claims. Gao et al. compare inline and post-hoc citation generation, and they evaluate citation quality through claim-level support and completeness Gao et al. (2023). Liu et al. study generative search engines from the reader’s point of view. They measure whether claims are supported by citations and whether users can verify those claims Liu et al. (2023). This work gives important standards for useful citations: important claims should be covered, cited sources should support them, and references should be easy to inspect. But these standards mainly compare the final claim with the final source. They do not test whether the source helped produce the claim. A model can answer from memory and then attach a passage that supports the answer. That citation may be correct, but it may still give a misleading account of provenance.

Citation faithfulness asks whether an emitted citation reflects the evidence that influenced generation. Wallat et al. define the difference between correctness and faithfulness in RAG and introduce adversarial interventions Wallat et al. (2025). Their method places answer-bearing strings in documents that were not the original source of support. If a model repeats the target statement and cites that inserted document, this is behavioral evidence of post-rationalization. Qi et al. study faithful attribution from inside the model by using context sensitivity and saliency to estimate which retrieved documents affect answer tokens Qi et al. (2024). These approaches answer different questions. Adversarial tests show behavior under controlled input changes. Internal attribution methods estimate influence in one forward pass.

The difference between retrieved evidence and stored model knowledge also appears in work on knowledge conflicts. PopQA studies when models answer from stored knowledge and when outside evidence helps, with entity popularity as an important factor Mallen et al. (2023). ConflictBank tests cases where supplied context conflicts with a model’s stored knowledge and measures which source the model follows Su, Zhang, Qu, Zhu, Li, Sun, Li, Zhang and Cheng (2024). Following the supplied context is related to citation faithfulness, but it is not the same thing. A context-matching answer shows that the model is sensitive to the supplied evidence. A faithful citation must also point to the document that actually supported the claim.

Mechanistic interpretability gives tools for testing causal ideas inside a model. Causal tracing and activation editing have found transformer components involved in factual recall, and they show that changing internal states can change model predictions Meng, Bau, Andonian and Belinkov (2022). TransformerLens provides hooks and activation caches that are commonly used for these patching experiments Nanda and Bloom (2022). The closest related work is by van Dort and Heuss, who study citation generation in Llama-3.1-8B-Instruct with activation patching on PopQA and a citation-token logit-difference metric van Dort and Heuss (2026). Their work shows how activation patching can be used to study citation-token behavior in a controlled setting. Our mechanistic experiment should be read as a replication-and-extension attempt rather than a new localization method: it applies a related citation-logit setup to a small PopQA capital-city sample, while the behavioral benchmark, conflict analysis, and hidden-state probes extend the evaluation beyond citation-token logits.

Linear probes ask a different question: can a simple classifier recover a target property from a model representation? Prior work has used such methods to study whether language-model activations contain information about truth and uncertainty. Burns et al. recover latent answers to true-or-false questions by finding directions in activation space that satisfy logical consistency constraints, even without labeled examples Burns, Ye, Klein and Steinhardt (2023). Azaria and Mitchell train classifiers on hidden activations to distinguish true from false statements, including statements generated by the model itself Azaria and Mitchell (2023). Li et al. train linear probes on attention-head activations to locate directions associated with truthful answers and then use those directions for inference-time intervention Li, Patel, Viégas, Pfister and Wattenberg (2023a).

These studies show that internal representations can support prediction of factual truth, latent knowledge, or semantic uncertainty. Our probe target is different: it is not whether an answer is factually correct or whether the model is uncertain, but whether a citation is post-rationalized under a controlled document intervention. This places citation faithfulness within the broader use of hidden-state probes for model monitoring while introducing an attribution-specific behavioral label. As with probing more generally, strong classification performance shows that information linked to the label is linearly available; it does not prove that the model uses the probed direction or that the direction causes the behavior Belinkov (2022). Probe evidence must therefore be interpreted separately from causal intervention evidence.

This distinction matters increasingly to information and computer science audiences. Model-internal attribution has been evaluated with context sensitivity and saliency Qi et al. (2024); knowledge-conflict benchmarks now quantify how supplied evidence competes with parametric memory Su et al. (2024); and mechanistic work examines citation-token behavior directly Wallat et al. (2025); van Dort and Heuss (2026). These studies provide complementary tools but do not jointly compare behavioral interventions, conflict sensitivity, causal patching, and representation-level prediction across several open-weight models. Building on them, this study connects the four methods in one reproducible pipeline and uses them to separate citation correctness from the stronger question of whether a citation reflects the evidence used during generation.

Complementary studies reinforce the importance of evaluating retrieval and evidence use as separate parts of an information system. Li et al. formulate conversational question answering as generative retrieval, showing that passage selection is itself a central modeling problem rather than a transparent precondition for answer generation Li, Yang, Wang, Wei and Li (2023b). Liu et al. likewise show that query generation and the management of conversational context affect legal-case retrieval, illustrating how changes in the information supplied to a model can alter the evidence available for an answer Liu, Wu, Zhang, Liu, Wang, Li, Zhang and Ma (2022). For evaluation, Wang et al. develop a scalable cross-language fact-checking study over 4,500 claims in nine languages and report substantial variation across languages and models Wang, Tsang and Zhou (2026). Together, these studies support a broader evaluation perspective: retrieval quality, contextual evidence selection, answer correctness, and the provenance of generated claims should be measured separately. Our study extends this perspective by testing whether the document named in a citation was behaviorally used, whether that signal is represented in hidden states, and whether targeted internal interventions can change it.

4. Datasets

The behavioral sample contains 1,444 unique Natural Questions identifiers and questions released with the experiments of Wallat et al. Wallat et al. (2025). The questions originate from Natural Questions Kwiatkowski et al. (2019); the retrieved evidence comes from the full KILT Wikipedia knowledge source Petroni et al. (2021). We converted Wikipedia pages into non-overlapping chunks of 100 whitespace tokens before BM25 retrieval and retained the top five chunks per query. Thus, Natural Questions defines the question sample and KILT defines the evidence population; the intervention counts in Table 2 are lower because they are conditional on an original cited statement and a constructible adversarial document.

The mechanistic sample came from the `akariasai/PopQA` test split Mallen et al. (2023). We retained capital-city relations and selected 31 clean/corrupted pairs using the citation-logit filter described below. The counterfactual sample came from the `CB_qa` training split associated with ConflictBank Su et al. (2024). We streamed that split, retained the first 2,000 examples with the expected schema, and sorted them by example identifier before generation; therefore, the ConflictBank results contain 2,000 examples per model. ConflictBank was used only for secondary validation, PopQA only for mechanistic analysis, and probe labels and features came from our behavioral interventions rather than either dataset.

All model weights and tokenizers came from their official Hugging Face releases. We used three instruction-tuned checkpoints: `meta-llama/Llama-3.1-8B-Instruct`, `Qwen/Qwen2.5-7B-Instruct`, and `google/gemma-3-12b-it`. These models were released by Meta, Alibaba's Qwen team, and Google DeepMind, respectively Meta AI (2024); Qwen Team (2024); Google DeepMind (2025). Llama 3.1 and Gemma 3 require license acceptance. Qwen2.5 is distributed under the Apache 2.0 license. Because these are open-weight models, we can generate answers, extract hidden states, and run activation-patching experiments. For the semantic-similarity baseline, we used `sentence-transformers/all-mpnet-base-v2` to encode document-query pairs, following the sentence-transformer bi-encoder approach Reimers and Gurevych (2019).

Dataset/source	Sample or coverage	Experimental role
Natural Questions Kwiatkowski et al. (2019)	1,444 unique questions	Main behavioral question sample
KILT Wikipedia Petroni et al. (2021)	Full knowledge source; 100-token chunks and top-5 retrieval	Retrieved evidence for the behavioral benchmark
PopQA Mallen et al. (2023)	31 clean/corrupted capital-city pairs	Llama-only mechanistic activation patching
ConflictBank Su et al. (2024)	2,000 examples per model	Secondary validation of context-parametric knowledge conflict

Table 1

Summary of datasets and their roles in the citation-faithfulness experiments.

5. Methodology

5.1. Research Architecture

The study has two complementary experimental paths. The main pipeline starts with Natural Questions, KILT Wikipedia, and ConflictBank; it retrieves top-5 BM25 contexts from 100-token Wikipedia chunks, runs deterministic generation with the three open-weight instruction-tuned models, applies behavioral interventions, and evaluates post-rationalization with citation labels and hidden-state probes. A separate auxiliary path uses PopQA capital-city pairs for Llama-only activation patching, component selection, and generation-level validation. Figure 1 summarizes this arrangement, which keeps the all-model behavioral comparison separate from the narrower Llama causal analysis.

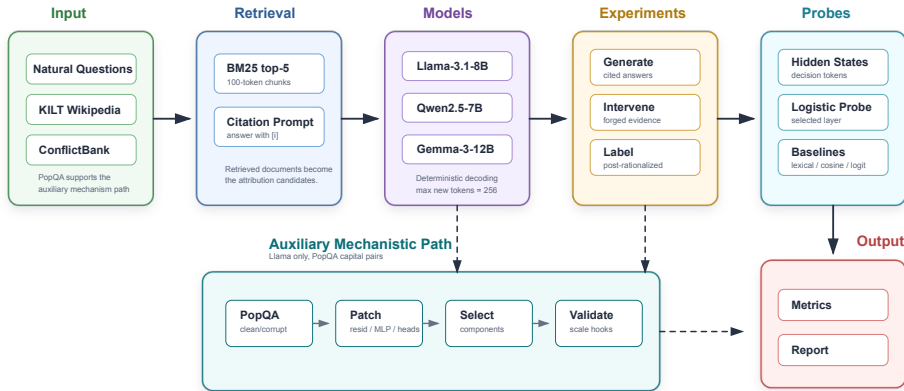


Figure 1: Research architecture for the citation-faithfulness experiments. The main path evaluates retrieval, generation, behavioral interventions, and probes across all three models. The auxiliary mechanistic path performs Llama-only activation patching on PopQA examples.

5.2. Task Setup

Each example begins with a question and a document collection. Let q_i be the question for example i , and let $D = \{d_1, \dots, d_N\}$ be the document corpus. The retriever ranks documents with a BM25 score $s(q_i, d)$ and returns the top- k documents:

$$R_k(q_i) = \text{TopK}_{d \in D} s(q_i, d), \quad (1)$$

In this study, $k = 5$. We write the retrieved set as $C_i = \{d_{i1}, \dots, d_{ik}\}$. The model receives the question and these retrieved documents as a prompt $x_i = (q_i, C_i)$ and generates an answer:

$$y_i = M_\theta(x_i), \quad (2)$$

Here, M_θ is one of the instruction-tuned models. The generated answer contains one or more statements and citation markers. For a target statement t_i and a cited document index c_i , citation correctness can be written as a support relation:

$$\text{Correct}(t_i, c_i) = \mathbb{1}\{d_{ic_i} \models t_i\}, \quad (3)$$

where $d_{ic_i} \models t_i$ means that the cited document supports the statement. This only checks the final text. It does not show whether the model used that document while producing the answer.

To test faithfulness, we change the retrieved context and observe the model's response. For each target statement, we create an intervention I_a that inserts an answer-bearing phrase into an adversarial document d_{ia} . The new retrieved context is

$$C_i^{(a)} = I_a(C_i), \quad (4)$$

and the model generates a new answer

$$y_i^{(a)} = M_\theta(q_i, C_i^{(a)}). \quad (5)$$

We then define two indicators. The first checks whether the target statement is recovered:

$$r_i^{(a)} = \mathbb{1}\{t_i \subset y_i^{(a)}\}. \quad (6)$$

The second checks whether the adversarial citation marker is used in the sentence that contains the recovered target phrase:

$$z_i^{(a)} = \mathbb{1}\{[a] \in \text{sent}(y_i^{(a)}, t_i)\}. \quad (7)$$

The post-rationalization label is therefore

$$\ell_i^{(a)} = r_i^{(a)} z_i^{(a)}. \quad (8)$$

A value of $\ell_i^{(a)} = 1$ means that the model recovered the target statement and cited the adversarial document. This is the main behavioral signal used in the study.

5.3. Models and Execution

We evaluated three open-weight instruction-tuned models: meta-llama/Llama-3.1-8B-Instruct, Qwen/Qwen2.5-7B-Instruct, and google/gemma-3-12b-it. Behavioral and probe experiments were run for all three models. Mechanistic activation patching was run only for Llama-3.1-8B-Instruct, which matches the mechanistic scope of the study. Generation used greedy decoding with a maximum of 256 new tokens. The experiments ran on Apple MPS with backend-selected reduced precision and no quantization.

We fixed the seed at 42 and used each model's official chat template, `do_sample=false`, one beam, and 256 new tokens. The system prompt told the model to use relevant documents, cite factual statements immediately with document numbers [1]–[5], cite only supporting documents, avoid citing a document merely because it shared an answer phrase, and answer from model knowledge without a citation when the documents lacked the answer. Each user prompt contained five numbered documents and one question. We saved the rendered prompt, input length, raw output, parsed citations, and a SHA-256 prompt hash for each generation.

For each original answer, we chose one statement with a single citation and a target phrase of at most eight tokens that appeared in its cited document. We tested three conditions: a random document that excluded the original pages and target phrase, a relevant retrieved document that the original answer did not cite, and another document cited elsewhere in that answer. We inserted the target phrase into the selected document. We labeled an output POST-RATIONALIZED only when it recovered the normalized target phrase and cited the altered document in the same statement.

5.4. Behavioral Post-Rationalization Benchmark

The main behavioral benchmark uses 1,444 Natural Questions queries and top-5 BM25 retrieval from KILT Wikipedia. For each model, we first generated answers with citations. The system instruction required citations for document-supported claims. We then parsed the cited statements and built interventions around target statements that the model originally cited.

Each target statement was tested under three adversarial conditions:

- **Random:** the answer-bearing phrase is inserted into an otherwise random document, placed at citation index [5].
- **Relevant uncited:** the phrase is inserted into a retrieved document that is topically relevant but was not the original cited source.
- **Cited other:** the phrase is inserted into another document that the model cited elsewhere in the original answer.

An intervention is available when we can construct the adversarial document. We label the output as POST-RATIONALIZED when two things happen: the model recovers the target statement, and the adversarial document’s citation marker appears in the sentence with the recovered target phrase. The main metric is the adversarial citation rate, conditioned on statement recovery and condition availability:

$$\text{PRR} = \frac{\#\{\text{recovered target and adversarial citation}\}}{\#\{\text{condition available and recovered target}\}}. \quad (9)$$

Wilson 95% intervals were computed for the conditional rate.

The original supporting document remains in the relevant-uncited and cited-other contexts. The random condition replaces document [5]; therefore, it also replaces the original supporting document when that document occupied slot [5] (18 Llama cases, 37 Qwen cases, and 113 Gemma cases). This position and source-retention difference limits the random condition as a strict control.

As a concrete example, Qwen2.5-7B-Instruct was asked, “When did Madras change its name to Chennai?” In the original run, it answered “Madras changed its name to Chennai in 1996 [5].” For the relevant-uncited intervention, the phrase “1996” was appended to retrieved document [1], a relevant Madras Railway passage that was not the original cited source. The model then produced “Madras changed its name to Chennai in 1996 [1].” Because the target statement was recovered and the newly modified document was cited in the same sentence, this example receives the label POST-RATIONALIZED. It illustrates the distinction between correctness and faithfulness: both documents can support the statement, but the intervention shows that the citation can shift toward a document after the answer phrase is placed in it. This single example does not establish the model’s internal causal process; the benchmark measures the observable behavior systematically across many examples.

5.5. ConflictBank Validation

We use ConflictBank as a secondary validation experiment, not as probe training data. ConflictBank examples contain a question, an original or parametric answer, a conflicting answer, and a piece of evidence that supports the conflicting answer. This lets us test what the model does when the retrieved context disagrees with what the model may already know.

For each example j , let q_j be the question, a_j^p be the original or parametric answer, a_j^c be the conflicting answer, and e_j^c be the conflicting evidence. Each model is run twice. First, it answers the question without any supporting document:

$$y_j^0 = M_\theta(q_j). \quad (10)$$

Second, it answers the same question with the conflicting evidence provided as the retrieved document:

$$y_j^c = M_\theta(q_j, e_j^c). \quad (11)$$

The generated answer is then assigned to one of three classes. After text normalization, it is labeled PARAMETRIC MATCH if it contains a_j^p but not a_j^c . It is labeled CONTEXT MATCH if it contains a_j^c but not a_j^p . It is labeled AMBIGUOUS if it contains both answers or neither answer:

$$g(y_j) = \begin{cases} \text{PARAMETRIC MATCH,} & a_j^p \subset y_j \text{ and } a_j^c \not\subset y_j, \\ \text{CONTEXT MATCH,} & a_j^c \subset y_j \text{ and } a_j^p \not\subset y_j, \\ \text{AMBIGUOUS,} & \text{otherwise.} \end{cases} \quad (12)$$

We apply this classifier to both y_j^0 and y_j^c . The main validation signal is the rate at which the with-context answer is labeled CONTEXT MATCH:

$$\text{CMR} = \frac{1}{n} \sum_{j=1}^n \mathbb{1}\{g(y_j^c) = \text{CONTEXT MATCH}\}. \quad (13)$$

In this study, $n = 2000$ for each model. A high context-match rate means that the model is strongly influenced by the supplied context when that context conflicts with stored knowledge. The resulting class counts are converted to percentages for the context-shift figure in Section 6.

5.6. Mechanistic Activation Patching

For mechanistic analysis, we built capital-city pairs from PopQA Mallen et al. (2023). A clean prompt contains a document of the form “A is the capital of S” and asks the matching question. A corrupted prompt replaces the clean document with a distractor document. The distractor is length-matched in tokenization for the subject and answer spans. We selected the first 31 examples where the citation logit difference is positive in the clean prompt and negative in the corrupted prompt:

$$\Delta = \text{logit}(\text{ "[")} - \text{logit}(\text{ ". "}). \quad (14)$$

For each selected example, we cached clean and corrupted activations. We then patched residual streams, MLP outputs, and attention heads. Denoising recovery and noising degradation were computed as

$$R_{\text{denoise}} = \frac{\Delta_{\text{patched}} - \Delta_{\text{corrupt}}}{\Delta_{\text{clean}} - \Delta_{\text{corrupt}}}, \quad D_{\text{noise}} = \frac{\Delta_{\text{clean}} - \Delta_{\text{patched}}}{\Delta_{\text{clean}} - \Delta_{\text{corrupt}}}. \quad (15)$$

We marked components as positive if their mean denoising recovery was at least 0.10. We marked components as necessary if their mean noising degradation was at least 0.20. We then tested the selected components by scaling them during generation on missed-citation and spurious-citation examples.

5.7. Linear Faithfulness Probes

For each model, we extracted hidden-state features from behavioral intervention examples labeled POST-RATIONALIZED or NOT-POST-RATIONALIZED. When an adversarial citation marker appeared, we used its first token as the decision position. Otherwise, we used the sentence position linked to the target phrase. For each layer, the feature vector was the mean hidden state over the decision token and the two answer tokens before it.

For layer λ and example i , let $h_{i,\lambda,t}$ be the hidden state at answer-token position t . If T_i is the decision token position, the probe feature is

$$u_{i,\lambda} = \frac{1}{3} \sum_{t=T_i-2}^{T_i} h_{i,\lambda,t}. \quad (16)$$

For each layer, the probe predicts the probability of post-rationalization with logistic regression:

Model	Condition	Post-rationalized	Available	Rate	Wilson 95% CI
Llama-3.1-8B	Random	0	274	0.0%	[0.0, 1.7]
Llama-3.1-8B	Relevant uncited	8	245	3.5%	[1.8, 6.7]
Llama-3.1-8B	Cited other	2	23	9.1%	[2.5, 27.8]
Qwen2.5-7B	Random	3	326	1.1%	[0.4, 3.2]
Qwen2.5-7B	Relevant uncited	27	299	9.7%	[6.7, 13.7]
Qwen2.5-7B	Cited other	11	59	21.6%	[12.5, 34.6]
Gemma-3-12B	Random	4	829	0.6%	[0.2, 1.5]
Gemma-3-12B	Relevant uncited	13	726	1.9%	[1.1, 3.2]
Gemma-3-12B	Cited other	18	117	17.8%	[11.6, 26.4]

Table 2

Behavioral post-rationalization rates conditional on condition availability and target statement recovery.

$$\hat{p}_{i,\lambda} = \sigma(w_{\lambda}^{\top} u_{i,\lambda} + b_{\lambda}), \quad (17)$$

where σ is the sigmoid function. The probe is trained with balanced binary cross-entropy,

$$\mathcal{L}_{\lambda} = - \sum_i \alpha_{\ell_i} [\ell_i \log \hat{p}_{i,\lambda} + (1 - \ell_i) \log(1 - \hat{p}_{i,\lambda})], \quad (18)$$

where ℓ_i is the post-rationalization label and α_{ℓ_i} is the class weight. This weighting is used because post-rationalized examples are much rarer than non-post-rationalized examples.

We split questions by `question_id` into 70% training, 15% validation, and 15% test sets. This avoids leakage across interventions for the same question. We trained one L2 logistic regression probe per layer with balanced class weights. We selected the layer with the highest validation AUROC and report test AUROC, AUPRC, and balanced accuracy in Table 4. The baselines used lexical overlap, citation-logit margin, and document-query cosine similarity from a sentence-transformer encoder. Lexical overlap was the Jaccard similarity between normalized target-statement and adversarial-document tokens; the citation-logit baseline was the marker-versus-period logit margin; and the semantic baseline was cosine similarity between question and adversarial-document embeddings from `sentence-transformers/all-mpnet-base-v2`. Baseline scores were evaluated on the same question-level test split as the selected-layer probes.

6. Experimental Results

6.1. Behavioral Post-Rationalization

The behavioral benchmark compares three types of forged documents: random documents, relevant documents that were not originally cited, and documents cited elsewhere in the original answer. Random forged documents are rarely cited, whereas rates are higher for relevant-uncited and cited-other documents, suggesting that models are more likely to cite forged evidence when it looks topically relevant or already fits the citation context. Table 2 reports the conditional post-rationalization rates.

The rates are conditional on different funnels rather than a common denominator. The original-answer files contain 1,444 generations per model; parseable cited answers numbered 629 for Llama, 916 for Qwen, and 1,106 for Gemma. Target extraction then yielded 274, 326, and 829 question-level intervention sets, respectively. Available cited-other cases were only 23, 59, and 117, so the cited-other percentages should not be treated as a definitive cross-model ranking.

6.2. ConflictBank

The ConflictBank experiment compares each model’s answer without supporting context with its answer after conflicting evidence is supplied. Each model therefore has two bars in Figure 2, with stacked colors showing whether the answer matches the model’s parametric answer, matches the supplied context, or remains ambiguous under the

Validation set	Count	Correction success rate
Missed citation	7	0.0%
Spurious citation	58	0.0%

Table 3

Mechanistic validation after component selection and scaling. Correction success rate is the percentage of examples whose citation behavior is fixed by the scaling intervention.

string-matching rule. We use this figure instead of a separate table because the main result is the change between the two settings. Without context, most outputs are ambiguous because the short matching rule often finds neither answer directly; with conflicting evidence, all three models shift toward the context-matching class. This is evidence of context sensitivity under the classifier, not a clean before–after measure of overriding parametric knowledge: the no-context condition was ambiguous for 1,994 Llama, 1,942 Qwen, and 1,916 Gemma examples. We therefore interpret the figure as a context-conditioned answer-class analysis rather than a definitive measure of knowledge conflict resolution.

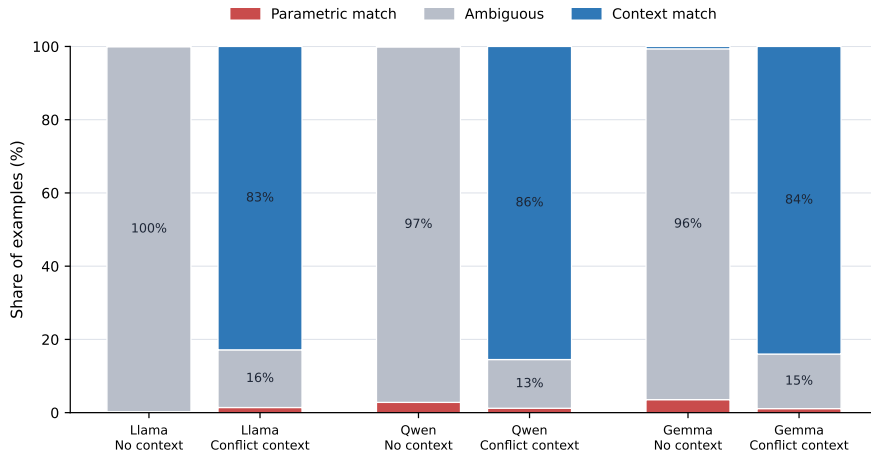


Figure 2: ConflictBank answer classes before and after conflicting evidence is supplied. The figure shows that all three models shift strongly toward the context-supported answer when conflict evidence is included.

6.3. Mechanistic Patching and Interventions

The PopQA mechanistic phase selected 31 clean/corrupted pairs that met the filtering rule. Patching produced 512 residual entries, 512 MLP entries, and 1,024 attention-head entries. Component selection found no positive citation heads or MLP components under the denoising threshold. It found one necessary attention head, layer 0 head 3, and 11 necessary MLP positions in layer 0. The full result is therefore better described as failed localization above the prespecified threshold than as identification of a citation mechanism. The citation-logit difference was measured at the final prompt position before generation, and the patching tables covered token positions 102–117.

Validation found 7 missed-citation examples and 58 spurious-citation examples. We define correction success rate as the percentage of validation examples whose citation behavior is fixed after component scaling. For missed-citation examples, success means that the intervention adds the expected citation while preserving the answer. For spurious-citation examples, success means that the intervention removes the unwanted citation while preserving the answer. Neither group had a successful correction: 0/7 missed-citation cases and 0/58 spurious-citation cases. Because no positive components passed the selection threshold, this result does not establish that a validated citation mechanism was resistant to scaling.

6.4. Faithfulness Probes

The probe datasets contained 15,328 Llama feature rows, 16,912 Qwen feature rows, and 70,272 Gemma feature rows. Table 4 reports test performance for the selected layer of each model. The selected layers fall around 60–68% relative depth. This suggests that the faithfulness signal is easiest to decode in middle-to-late residual streams. These

Model	Layer	Rel. depth	AUROC	AUPRC	Bal. acc.	Highest baseline AUROC
Llama-3.1-8B	21	0.68	0.943	0.200	0.479	0.757
Qwen2.5-7B	17	0.63	0.969	0.325	0.918	0.908
Gemma-3-12B	28	0.60	0.831	0.357	0.616	0.716

Table 4

Test performance of selected-layer linear probes. Highest baseline AUROC is the highest score among lexical overlap, citation-logit margin, and document-query cosine similarity.

row counts include all extracted layers and are not independent examples. At the selected layer, the test sets contained 71 Llama rows with 1 positive label, 100 Qwen rows with 2 positives, and 226 Gemma rows with 4 positives; the corresponding train/validation positive counts were 7/2, 32/7, and 28/3. The probe results should therefore be read as small, question-level evaluations rather than large independent samples.

The probes have higher AUROC than the best listed baseline for all three models. Qwen has the highest balanced accuracy among the three models. Llama has high AUROC but lower balanced accuracy at the default 0.5 threshold, mainly because it has very few positive test examples. Gemma has lower AUROC than Llama and Qwen, while its AUPRC is higher than the other two models under the observed label imbalance. To quantify uncertainty, we resampled complete question groups 1,000 times and computed percentile 95% intervals. The resulting AUROC intervals were [0.877, 0.973] for Llama, [0.928, 1.000] for Qwen, and [0.613, 1.000] for Gemma; the corresponding AUPRC intervals were [0.125, 0.500], [0.143, 1.000], and [0.016, 1.000]. Behavioral rates use Wilson 95% intervals.

7. Discussion

The behavioral results show that citation faithfulness is not simply present or absent. Although the models do not blindly cite arbitrary forged documents, they are more likely to cite adversarial documents when those documents are topically relevant or already plausible in the citation context, suggesting that citation selection can respond to document-answer compatibility even when the document is not the true source of the answer.

The ConflictBank results point in the same direction from another angle: when supplied evidence conflicts with stored model knowledge, all three models usually follow the supplied context. This sensitivity is useful when the context is correct, but it also means that document placement and surface evidence can strongly steer answers.

The mechanistic results are more mixed. Activation patching identifies early necessary components, but the later scaling intervention does not successfully control behavior. One possible reason is that citation decisions are distributed across many parts of the model and depend on context. Components that affect a next-token citation logit in a controlled prompt may not be enough to control a full multi-token answer. Another possibility is that local MPS precision and the simple scaling method made the validation too weak. Either way, the negative validation limits how strong the mechanistic claim can be.

The probe results are practically useful because, even with simple logistic regression, hidden states contain a signal that predicts post-rationalization. Although this does not prove that the probed feature causes unfaithful citation behavior, it suggests that faithfulness risk could be monitored without rerunning the full adversarial intervention for every generation.

7.1. Theoretical and Practical Implications

Theoretically, the findings refine attribution in RAG by separating three levels of evidence. First, citation correctness is a textual support relation between a claim and a document. Second, behavioral faithfulness is tested by changing the document’s evidential role and observing whether the model changes its answer or citation. Third, mechanistic faithfulness would require identifying internal states whose intervention reliably changes generation. The observed post-rationalization rates establish that correctness and behavioral faithfulness can diverge. The probe results further show that this divergence is represented in hidden states, while the failed scaling intervention cautions against interpreting decodability as causal control. This distinction extends recent attribution and knowledge-conflict research by placing support, sensitivity, prediction, and intervention on separate evidential levels.

Practically, RAG evaluations should report citation correctness together with intervention-based faithfulness tests. Systems should not treat a relevant cited document as proof that the document caused the answer. The results also

motivate a two-stage monitoring design: use calibrated hidden-state probes as a low-cost screening signal, then apply document-level verification or controlled re-generation to flagged cases. The low availability of the cited-other condition and the imbalance in probe labels show that deployment thresholds must be calibrated on an independent sample. Finally, citation interfaces should distinguish “supports the claim” from “was used to generate the claim” unless the latter has been directly tested.

8. Limitations

This study uses one specific definition of citation faithfulness. It detects post-rationalization under answer-string insertion interventions. It does not cover every possible kind of unfaithful attribution. The Natural Questions setup depends on BM25 retrieval over KILT Wikipedia chunks and on parser rules for cited statements. The cited-other condition has low availability for Llama, so its confidence interval is wide. The mechanistic experiment is also narrow: it uses only Llama, only PopQA capital facts, and a citation-token-versus-period logit difference as the target. The probe labels are imbalanced, especially for Llama and Gemma, so ranking metrics should be interpreted alongside threshold-dependent metrics. Finally, the local run used Apple MPS rather than CUDA, so exact numerical results may differ from the original CUDA-oriented setup.

The behavioral design did not include no-insertion or placebo-insertion arms, nor did it randomize document position; random forged documents were fixed at slot [5]. The reported rates therefore measure behavior under answer-string insertion and should not be interpreted as an isolated insertion effect. We also did not manually audit approximately 100 labels per model or test cross-condition probe transfer, so substring and sentence parsing errors and intervention-specific probe signals remain possible. Finally, Gemma-3-12B is larger than the two other models, and the mechanistic sample is limited to one model and one relation type.

9. Conclusion

This study investigated citation faithfulness in open-weight retrieval-augmented generation by distinguishing textual support from actual evidence use. A citation is correct when its document supports the generated claim, but it is faithful only when it reflects the evidence that influenced generation. We examined this distinction across multiple instruction-tuned model families through behavioral document interventions, knowledge-conflict evaluation, mechanistic activation patching, and hidden-state probing. The behavioral experiments showed that models rarely cited random forged documents, but cited them more often when they appeared relevant or plausible. The knowledge-conflict evaluation further showed that retrieved context can influence outputs when it disagrees with stored knowledge. Together, these findings indicate that models can rely on supplied evidence while also attaching plausible citations that do not necessarily identify the source responsible for a claim. Document-answer compatibility, contextual influence, and faithful provenance are therefore related but distinct properties.

The internal analyses clarified both what can be detected and what can be controlled. Activation patching showed that an early attention head and several early MLP positions affected the model’s score for generating a citation token. However, scaling these components during full generation did not add missing citations or remove unwanted citations while preserving the answer. This means that changing a next-token citation score was not sufficient to control citation behavior across a complete generated response. Linear probes distinguished post-rationalized from non-post-rationalized outputs more effectively than the listed surface-level baselines by AUROC, indicating that hidden states contain information associated with citation-faithfulness risk; however, this establishes predictability rather than causality and would require calibration before deployment. Taken together, the experiments indicate that citation faithfulness cannot be represented by a single metric: support-based evaluation checks whether citations justify claims, controlled interventions test evidence reliance, conflict tests characterize sensitivity to context, mechanistic methods examine causal structure, and probes assess whether risk is internally detectable. RAG systems should therefore connect claims not only to documents that support them but also, where possible, to evidence that influenced their generation.

Future work should test whether these findings generalize across retrieval methods, knowledge domains, model sizes, model families, and citation formats. Mechanistic studies should use more precise, position-aware interventions that target the selected activations without changing unrelated parts of the answer, and should evaluate whether effects remain stable across prompts and longer generations. Probe-based monitoring should be tested on independent datasets, calibrated under realistic class imbalance, and evaluated at decision thresholds suited to deployment rather than only

by ranking metrics. Combining stronger causal tests with calibrated monitoring could support systems that detect unfaithful citations during generation and correct them without changing the factual content of the answer.

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